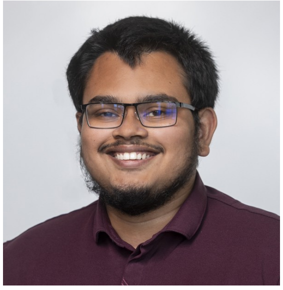
Power System Engineer

MD MAHBUB ALI



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Professional Summary

Electrical Engineering graduate from Aalborg University with a specialization in Electrical Power Systems and High Voltage, experienced in modelling and analysing inverter-dominated power systems through advanced academic simulation projects. Skilled in PSCAD and MATLAB for studying grid integration, inverter behaviour, and system dynamics. Confident enough to work in international teams, produce clear technical documentation, and present analytical findings to diverse audiences.

Skills

Simulation & Analysis
PSCAD
MATLAB Simulink
DIgSILENT

Power Systems
Power system analysis
Grid-forming control
Grid Code
Inverter-dominated stability

Programming & Data
MATLAB
LaTeX/Overleaf
Python

Languages
English (Intermediate)
Danish (Beginner)

Referrals

Sanjay Chaudhary
Associate Professor,
AAU Energy
skc@energy.aau.dk

Jakob Mænnchen
Engineering Management,
Siemens Gamesa
linkedin.com/in/jakobmaennchen

Education

Master of Science (MSc) in Engineering (Energy Engineering)
2023–2025
Aalborg University, Denmark

- Focused specialization in power system fundamentals, advanced power electronics, wind energy system & control, and battery energy system technologies.
- Developed a collaborative mindset through multinational project teamwork, including technical documentation, reporting, and presenting technical findings.

Selected Academic Projects

Synchronization & Fault Analysis of GFM Inverter (PSCAD)
2024–2025

- Modelled synchronization between a grid-forming VSG and a synchronous generator and analysed dual-VSG islanded operation.
- Simulated three-phase fault events to observe current stress and dynamic behaviour under short-circuit disturbances.

Black-Start Service using GFM Control System (PSCAD) 2024

- Simulated an 80 MVA system supplying a 20 MW / 15 MVAr load in a black-start scenario.
- Configured droop settings and control parameters using operational data from an industry partner.
- Evaluated voltage and frequency recovery and highlighted operating limits for reliable restart.

ML-Assisted Power Flow Calculation with Graph Neural Networks (DIgSILENT & Python) 2024

- Explored the use of GNN models to approximate Newton–Raphson power flow solutions.
- Tested performance and accuracy on sample PowerFactory cases and compared classical and ML-supported approaches and discussed potential use in future grid analytics.